

PS 3 · WHEN EVERY SECOND COUNTS

RecoveryLens

Early warning for the weeks after a stroke —
when the patient is out of the hospital's sight.

19,435

patients trained on

3,035

external validation

₹7

per patient, 6 months

MedInnovate 2026 · Grand Finale

Akash Chandrasekaran

The emergency starts where nobody is looking

1.29M

strokes in India each year

GBD 2019

699,000

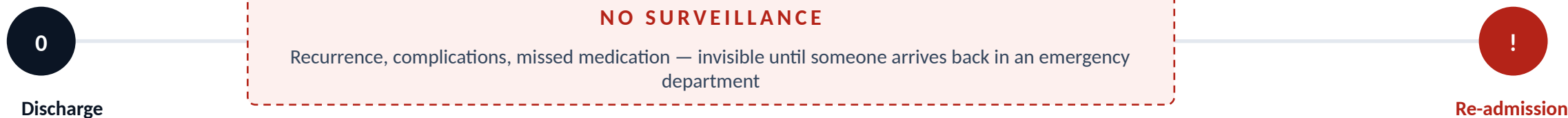
stroke deaths each year

GBD 2019

1 in 2

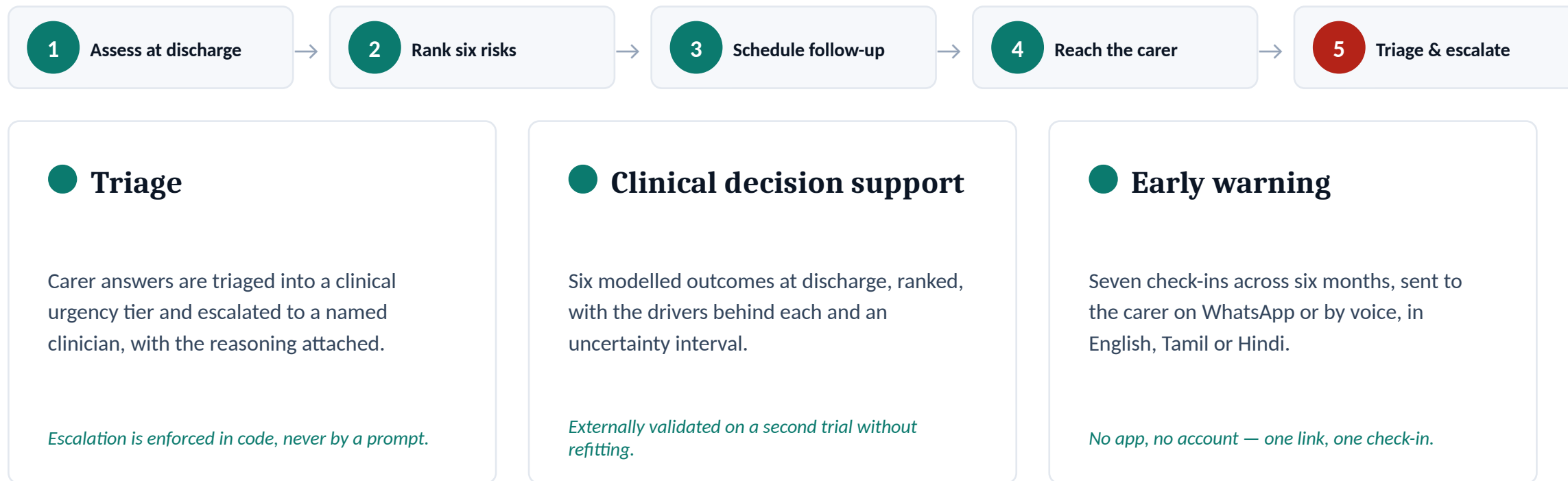
dead or dependent at 6 months

IST-1, our training cohort



In a resource-constrained setting the follow-up clinic is weeks away, transport is expensive, and the person who would notice a problem first — the family carer at home — has no way to tell anyone and no idea what to look for.

PS 3 asks for three things. We built all three.



We are not competing with the ambulance. We extend emergency capability into the one window with no monitoring at all — where catching a deterioration early is what prevents the emergency.

LIVE DEMONSTRATION

Discharge to escalation, in under three minutes

1

Paste a discharge summary

21 fields fill themselves, each carrying the sentence it came from

2

Review and submit

Six ranked outcomes, drivers, and a follow-up schedule with citations

3

Send the day-7 check-in

Real WhatsApp message, in Tamil, to a real phone

4

Answer it badly

Triage escalates, and the clinician's inbox shows why

What is real

- Everything you are about to see is running, not mocked
- Real models, real guideline text, real Twilio delivery
- Not built: clinical validation, hospital integration
- Synthetic patient — no real identifiers anywhere

What is actually built

● Discharge autofill

Paste the summary or upload the PDF. 21 fields extracted, each carrying the verbatim sentence it came from.

Claude with schema-constrained output

● Risk engine

Six outcomes scored and ranked, with SHAP drivers and split-conformal uncertainty intervals per patient.

scikit-learn · 4 MB, no GPU

● Cited guidance

723 guideline passages retrieved and quoted verbatim with section numbers. Never generated, never paraphrased.

TF-IDF + embeddings, measured refusal floor

● Follow-up scheduler

Seven dates across six months, each labelled guideline-backed or our own operational choice.

APScheduler · deterministic rules

● Multilingual messaging

WhatsApp and SMS in English, Tamil and Hindi. Consent gate, STOP keyword, rate limiting.

Twilio · back-translation verified

● Voice check-ins

Carers answer by speaking. Transcript is read back and must be confirmed before it is recorded.

Speech-to-text + text-to-speech

● Triage agent

An LLM with tools reads the carer's answer and assigns urgency. Escalation is monotonic and enforced after it speaks.

Tool use · code-enforced ceiling

● Clinician workspace

Patient records, escalation inbox, organisation-scoped accounts, consent and opt-out controls with an audit trail.

React + TypeScript · session auth

19,435

patients trained on

0.802 → 0.790

held out → external, IST-3

723

cited guideline passages

31

API endpoints

460

automated tests

Every model-backed feature degrades to a deterministic rule rather than failing: if the LLM is unreachable, guidance still resolves, triage still escalates, and the form still works. Nothing on this slide has a single point of failure that reaches a patient.

₹7 buys six months of watching one patient

≈ ₹7

per patient, for six months of follow-up

- 7 check-ins, sent and answered
- Meta utility message ₹0.115 each
- Twilio \$0.005 per message
- One small VM, no GPU
- No hospital IT integration to start

What it changes



Deterioration found in days

A question in front of the carer on days 3, 7, 14, 30, 42, 90 and 180.



The carer stops guessing

Told what to watch for, in their language, with guideline text behind it.



Costs a clinician nothing until it needs them

Only escalations reach the inbox. Silence costs no one any time.

What we do not claim



Training data is from 1991–96

Predates routine thrombectomy. The ranking transports; absolute risk would not.



No Indian cohort, no prospective test

Nothing here has been validated on Indian patients or tested on anyone.



Autofill tested on summaries we wrote

Perfect precision on five synthetic summaries is a small claim.

A single avoided re-admission pays for roughly ten thousand patients' worth of follow-up.

We have not measured that avoidance — that is what a clinical study would have to establish. Offered as an order of magnitude, not a result.

THE TAKEAWAY

The window after discharge is the cheapest place to prevent an emergency — and the only place nobody is watching.

RecoveryLens puts a question in front of the family carer on the days that matter, understands the answer in their language, and wakes a clinician only when it should.

Working today

Assessment, models, guidance, messaging, voice, triage, escalation

Next

Clinical review, an Indian cohort, a prospective pilot on one ward